Variable Selection
in Linear Models

Subset Selection · Shrinkage Methods · Model Evaluation

Based on: An Introduction to Statistical Learning (Chapter 6)

Agenda

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The problem with RSS/R^2 for model selection

02 **Subset Selection**

Best subset, forward, backward & hybrid

03 **Choosing the Best Model**

Cp, AIC, BIC, Adjusted R^2, Cross-Validation

04 **Shrinkage Methods**

Ridge Regression & The Lasso

05 **Tuning Parameter Selection**

Cross-validation for λ

Why Variable Selection?

The Problem

Including too many predictors hurts model quality:

- RSS always decreases as predictors are added
- R^2 always increases with more variables
- Training error ≠ Test error
- Overfitting leads to high variance
- Hard to interpret large models

Goals of Variable Selection

Find a subset of predictors that:

- Minimizes test (not training) error
- Reduces model complexity
- Improves interpretability
- Reduces variance at cost of small bias
- Selects signal variables, removes noise

Three Approaches to Variable Selection

Subset Selection	Shrinkage Methods	Dimension Reduction
Best Subset	Ridge Regression	Principal Components
Forward Stepwise	The Lasso	Partial Least Squares
Backward Stepwise	Shrink coefficients → 0	Project to subspace
Hybrid Methods	Retain all predictors (Ridge)	(Beyond this chapter)

Focus of this presentation: Subset Selection + Shrinkage Methods

Part 1: Subset Selection

Choosing the right predictors from 2^p possible models

Best Subset Selection

Fit a separate least squares regression for every possible combination of predictors.

Step 1

Null Model M_0

Model with no predictors — predicts sample mean for every observation.

Step 2

Fit All Size-k Models

For k = 1, 2, …, p: fit all C(p,k) models with exactly k predictors.
Pick the best M_k = smallest RSS (or largest R^2).

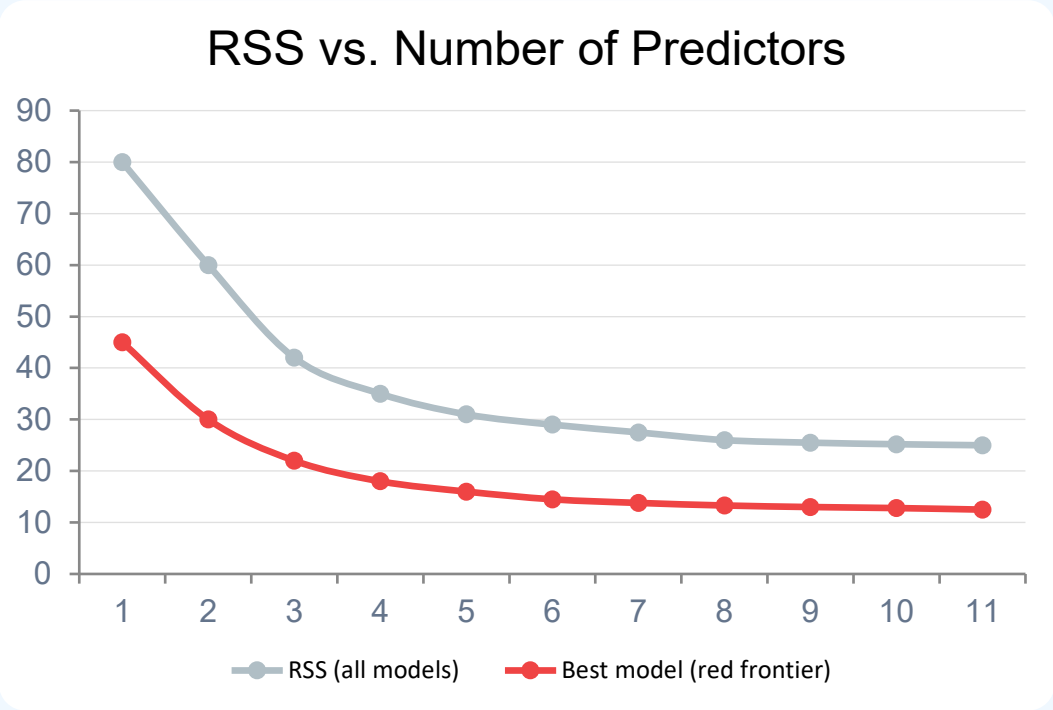
Step 3

Select Single Best Model

Choose from M_0, M_1, …, M_p using: Validation Error, C_p (AIC), BIC, or Adjusted R^2.

⚠️ **Computational limit: 2^p models grow exponentially — infeasible for p > 40**

Best Subset: RSS and R^2 Behavior



Red frontier: best model for each size k

RSS Monotonically ↓

Adding predictors always reduces RSS — can't use it to pick the best model.

R^2 Monotonically ↑

R^2 always rises with predictors — models with all variables win trivially.

Need Test Error

Use validation error, C_p, BIC, or adjusted R^2 to select model size.

Forward Stepwise Selection

Starts with null model, adds one predictor at a time — the one giving greatest fit improvement.



Best model selected using: C_p, AIC, BIC, Adjusted R^2, or Cross-Validation

Computational Comparison

Method	Models Fitted (p=20)	Works when n < p?
Best Subset	1,048,576 (2^{20})	No
Forward Stepwise	211	Yes ✓
Backward Stepwise	211	No

Backward Stepwise Selection

Starts with the full model, removes the least useful predictor one at a time.

1

Start: Full Model M_p

Begin with all p predictors in the model.

2a

Consider Removing One

For each iteration k = p, p-1, …, 1: consider all k models that omit exactly one predictor from M_k.

2b

Choose Best M_{k-1}

Pick the model with the smallest RSS (or highest R^2) — this is M_{k-1}.

3

Select Final Model

Use validation error, C_p, BIC, or adjusted R^2 to choose from M_0, M_1, …, M_p.

Key Constraint: requires n > p to fit the initial full model (use Forward Stepwise when n < p)

Best Subset vs. Forward Stepwise: Credit Data

Both methods often agree — but not always. Forward stepwise cannot revisit earlier choices.

# Variables	Best Subset	Forward Stepwise
1	rating	rating
2	rating, income	rating, income
3	rating, income, student	rating, income, student
4	cards, income, student, limit	rating, income, student, limit

✓ Match ✗ Differ

Forward stepwise keeps 'rating' in the 4-variable model because M_1 already contains it.
Best subset can replace it — finding globally optimal subsets at each size.

Hybrid Approaches

Forward Stepwise

Start empty, add predictors one by one. Greedy — cannot remove what was added.

Backward Stepwise

Start full, remove predictors one by one. Requires n > p.

Hybrid

Add predictors sequentially BUT also remove variables that no longer improve fit after each addition.

Hybrid methods mimic best subset selection more closely while maintaining the computational efficiency of forward/backward stepwise.

Part 2: Choosing the Optimal Model

Estimating test error to select among candidate models

C_p and AIC Criteria

Indirectly estimate test error by adjusting training RSS for model complexity.

Mallow's C_p

$$C_p = (1/n) \cdot [RSS + 2d \cdot \hat{\sigma}^2]$$

- d = number of predictors, $\hat{\sigma}^2$ = estimated error variance
- Penalty $2d\hat{\sigma}^2$ increases with more predictors
- Select model with LOWEST C_p

AIC (Akaike Information Criterion)

$$AIC = (1/n) \cdot [RSS + 2d \cdot \hat{\sigma}^2]$$

- For Gaussian errors: AIC $\propto C_p$
- Both are equivalent for least squares — only C_p shown
- Defined for a broader class of models (max likelihood)

Credit data example: C_p selects 6-variable model with predictors: income, limit, rating, cards, age, student

BIC — Bayesian Information Criterion

$$BIC = (1/n) \cdot [RSS + \log(n) \cdot d \cdot \hat{\sigma}^2]$$

C_p vs. BIC: Key Difference

Criterion	Penalty Term	Tendency	Credit Data
C_p / AIC	$2d \cdot \hat{\sigma}^2$	Moderate model size	6-variable model
BIC	$\log(n) \cdot d \cdot \hat{\sigma}^2$	Favors smaller models	4-variable model

Why BIC prefers smaller models:

$\log(n) > 2$ for $n > 7$ → BIC penalizes additional variables more heavily than C_p.

BIC is derived from a Bayesian perspective — selects the model most likely to be the true model.

Adjusted R^2

$$\text{Adjusted } R^2 = 1 - [RSS / (n - d - 1)] / [TSS / (n - 1)]$$

How it works

↑

Select HIGHEST value

Unlike C_p, AIC, BIC — a larger adjusted R^2 indicates a better model.

⚠

Less theoretical backing

Popular and intuitive, but C_p, AIC, BIC have stronger asymptotic theoretical justification.

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Key term: RSS/(n−d−1)

Dividing by (n−d−1) penalizes adding unnecessary variables. As d grows, denominator shrinks, so RSS/(n−d−1) may increase even if RSS decreases.

📊

Credit data result

Adjusted R^2 selects a 7-variable model — one more than C_p/AIC.

Validation & Cross-Validation

Direct estimation of test error — no assumptions about the true model required.

Validation Set

1. Split data into training set (e.g. 75%) and validation set (25%)
2. Fit all candidate models on training set
3. Evaluate each model on validation set
4. Pick model with lowest validation error

Simple, but results depend on random split

k-Fold Cross-Validation

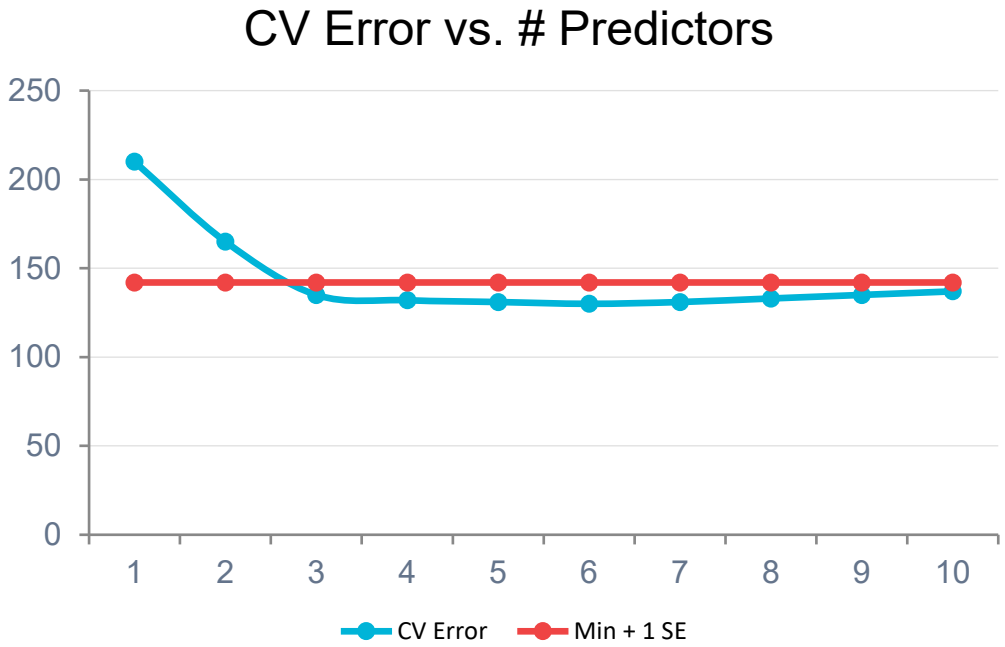
1. Divide data into k equal folds (e.g. k = 10)
2. For each fold: train on remaining k−1, test on the fold
3. Average errors across all k folds for each model size
4. Pick model size with lowest average CV error

More stable — recommended approach today

Credit data: Both validation set and 10-fold CV select a 6-variable model.
Advantage: direct test error estimate, no need for $\hat{\sigma}^2$ or model degrees of freedom.

The One-Standard-Error Rule

When several models have similar estimated test error — prefer parsimony.



Rule:

Select the SMALLEST model whose estimated test error is within 1 standard error of the lowest point on the curve.

Rationale:

If models look roughly equivalent, choose the simplest (fewest predictors).

Credit data result:

1-SE rule selects 3-variable model, even though 6-variable has lowest CV error.

Part 3: Shrinkage Methods

Constrain coefficient estimates toward zero to reduce variance

Shrinkage Methods: Core Idea

Instead of selecting a subset, fit a model using ALL p predictors — but constrain the coefficient estimates.

Least Squares → Minimize:

$$\text{RSS} = \sum (y_i - \beta_0 - \sum_j \beta_j x_{ij})^2$$

Shrinkage → Minimize RSS + Penalty on β:

Ridge Regression	The Lasso
$\text{RSS} + \lambda \cdot \sum_j \beta_j^2 \quad (\ell_2 \text{ penalty})$	$\text{RSS} + \lambda \cdot \sum_j \lvert\beta_j\rvert \quad (\ell_1 \text{ penalty})$
Shrinks all coeff toward 0; none set exactly to 0	Shrinks coeff toward 0; some set EXACTLY to 0 → variable selection

Ridge Regression

$$\min_{\beta} \{ \Sigma(y_i - \beta_0 - \Sigma_j \beta_j x_{ij})^2 + \lambda \cdot \Sigma_j \beta_j^2 \} \leftarrow \text{shrinkage penalty}$$

$\lambda = 0$

Penalty disappears → identical to ordinary least squares. High variance, no bias.

$\lambda \to \infty$

All coefficients shrink to zero → null model (predicts mean). High bias, zero variance.

Optimal λ

Intermediate λ balances bias-variance tradeoff. Chosen via cross-validation.

Standardize!

Predictors must be standardized before ridge regression — penalties depend on variable scale.

Intercept β_0

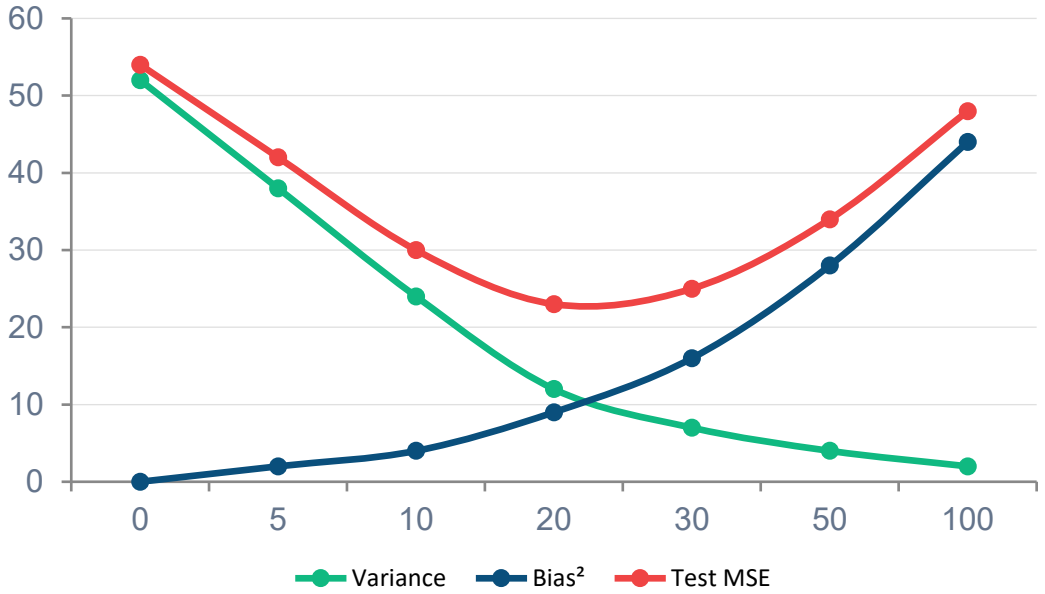
The intercept is NOT penalized — we only shrink $\beta_1, \ldots, \beta_p$ toward zero.

All p vars

Ridge keeps all p predictors — shrinks toward zero but never exactly to zero.

Ridge: Bias-Variance Tradeoff

As λ increases: flexibility decreases → variance decreases, bias increases.



Variance (Green)

Decreases rapidly as λ increases. Small λ = high variance.

Bias² (Blue)

Near 0 when λ=0 (OLS). Rises as coefficients are forced toward 0.

Test MSE (Red)

= Variance + Bias². Minimized at intermediate λ ≈ 30 in this example.

The Lasso

$$\min_{\beta} \{ \Sigma(y_i - \beta_0 - \Sigma_j \beta_j x_{ij})^2 + \lambda \cdot \Sigma_j |\beta_j| \} \leftarrow \ell_1 \text{ penalty}$$

Key Properties

Automatic Variable Selection

The ℓ_1 penalty forces some coefficients to EXACTLY zero → sparse models, like best subset selection.

Interpretability

Resulting models involve only a subset of predictors — far easier to interpret than Ridge.

Sparse Models

When λ is sufficiently large, many $\beta_j = 0$. Only 'signal' variables retain non-zero coefficients.

Tuning Parameter

$\lambda = 0$: same as OLS. $\lambda \to \infty$: all coefficients zero. Cross-validation selects optimal λ.

Why Lasso Performs Variable Selection

The geometry of the ℓ_1 constraint causes coefficients to hit zero — ℓ_2 does not.

Lasso (ℓ_1) Constraint — $|\beta_1| + |\beta_2| \leq s$

Diamond shape — has CORNERS at each axis.
Ellipses from OLS contours hit the corner → coefficient = 0 → variable excluded.

Ridge (ℓ_2) Constraint — $\beta_1^2 + \beta_2^2 \leq s$

Circular shape — NO corners.
Ellipses intersect the circle at a generic point → coefficients are non-zero (just small).

In higher dimensions ($p > 2$): Lasso constraint = polytope (sharp edges) → sparse solutions.

Ridge vs. Lasso: Side-by-Side

Property	Ridge Regression	The Lasso
Penalty	$\lambda \cdot \Sigma \beta_j^2$ (ℓ_2 norm)	$\lambda \cdot \Sigma \lvert\beta_j\rvert$ (ℓ_1 norm)
Coefficients	Shrunk toward 0, never exactly 0	Some set exactly to 0
Variable selection	No — includes all p predictors	Yes — automatic selection
Constraint region	Smooth circle (no corners)	Diamond with corners
Interpretability	Lower (all vars retained)	Higher (sparse model)
Best when…	Many predictors, all with similar effect	Few true signal predictors
Prediction accuracy	Better when all $\beta \neq 0$	Better when many $\beta = 0$
Shrinkage type	Proportional shrinkage	Soft-thresholding

Cross-validation determines which method works better on a given dataset.

Shrinkage Mechanics: A Simple Special Case

When X is diagonal (n = p), the shrinkage formulas have explicit, intuitive solutions.

Ridge Regression

$$\hat{\beta}^R_j = y_j \,/\, (1 + \lambda)$$

Each coefficient is simply scaled down by factor 1/(1+λ). All coefficients shrunk proportionally — none reach zero.

The Lasso

$$\hat{\beta}^L_j = \text{sign}(y_j) \cdot \max(|y_j| - \lambda/2,\ 0)$$

Coefficients with $|y_j| \le \lambda/2$ are set exactly to zero. Others are shrunk by a constant λ/2. This is SOFT-THRESHOLDING.

Key: Ridge shrinks ALL coefficients proportionally. Lasso shrinks by a fixed amount — setting small ones to 0.

Bayesian Interpretation

Ridge and Lasso emerge naturally from Bayesian regression with different priors on β.

Ridge Regression

Gaussian prior on β

$$p(\beta) \propto \exp(-\lambda \cdot \Sigma \beta_j^2)$$

Normal (Gaussian) distribution, flat at zero.

Ridge solution = Posterior MODE (and mean).

Expects: coefficients distributed randomly around zero.

The Lasso

Double-Exponential (Laplace) prior on β

$$p(\beta) \propto \exp(-\lambda \cdot \Sigma |\beta_j|)$$

Laplace distribution — steeply peaked at zero.

Lasso solution = Posterior MODE only.

Expects: many coefficients are EXACTLY zero a priori.

Both methods assume: linear model + normal errors + simple β prior → leads to penalized likelihood estimation.

Part 4: Selecting the Tuning Parameter λ

Cross-validation for optimal regularization strength

Selecting λ via Cross-Validation

The tuning parameter λ controls the strength of regularization — must be chosen carefully.

1

Define a grid of λ values

Typically spanning from very small (≈ OLS) to very large (null model), e.g. 0.001 to 10,000 on log scale.

2

Compute cross-validation error

For each λ in the grid, compute k-fold CV error across all candidate models.

3

Select optimal λ

Choose the λ that yields the smallest cross-validation error.

4

Refit on full data

Using the selected λ, refit the model on all available observations to get final coefficient estimates.

Credit Data Results:

Ridge: optimal λ is small (near OLS) — mild shrinkage only. | Lasso: finds correct 2 signal vars out of 45 in simulated example.

Practical Guide: When to Use Which Method

p is small (< 40)

Best Subset · Forward/Backward Stepwise

p is large (≥ 40) or n < p

Forward Stepwise · Ridge Regression · Lasso

Many predictors, all roughly equally important

Ridge Regression

Few true signal predictors (sparse model expected)

Lasso

Model interpretability is critical

Lasso · Best Subset

Unsure which method works best

Use cross-validation to compare · Both Ridge and Lasso · Evaluate on validation data

Key Takeaways

1

RSS/R^2 are insufficient

They always improve with more predictors — use test error estimates (C_p, BIC, adjusted R^2, CV) instead.

2

Best Subset is thorough but slow

Computationally infeasible for $p > 40$. Forward/backward stepwise offer efficient alternatives with $O(p^2)$ models.

3

Cross-validation is king

Provides direct, assumption-free test error estimates. The one-standard-error rule favors simpler models when errors are similar.

4

Ridge reduces variance

By shrinking coefficients toward zero — best when all predictors are weakly related to the response.

5

Lasso selects variables

ℓ_1 penalty forces exact zeros — best when only a few predictors truly matter. Interpretable sparse models.

6

No universal winner

Neither Ridge nor Lasso always dominates. Cross-validate to determine which works best for your data.

Based on: James, Witten, Hastie & Tibshirani — An Introduction to Statistical Learning, Chapter 6